

Computer Network (Numerical)

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2017-Spring

1) 192.168.10.0/27, calculate;

i) No. of subnets

N	Subnet	Hosts
---	--------	-------

ii) No. of hosts per subnet.

80 Here, 27 means that the subnet mask has 27 bits 1s and 5 bits 0s. So in binary mode our subnet mask is:

11111111. 11111111. 11111111. 11100000

(First 27 bits are 1s and 5 bits are 0s).

So there is 3 subnetting bit.

And,

∴ The IP address in binary mode is:

11000000. 10101000. 00001010. 00000000

After binary AND, IP with mask, we get:-

11000000. 10101000. 00001010. 00000000

∴ Network Address = 192.168.10.0

∴ Subnet mask = 255.255.255.224

∴ No. of subnets = $2^{\text{no. of subnet bits}} = 2^3 = 8$ Ans

∴ No. of hosts per subnet = $2^{\text{no. of host bits}} - 2 = 2^5 - 2$
= 30 Ans

7-9-8

∴ valid subnet = $256 - 224 = 32$

i.e. block size of 32. starting from zero it is 0, 32, 64, 96, ~~128~~, 160, 192, 224, ~~256~~

∴ Broad Cast Address :

Subnet Address :

192.168.10.31

192.168.10.0

192.168.10.63

192.168.10.32

192.168.10.95

192.168.10.64

192.168.10.127

192.168.10.96

192.168.10.159

192.168.10.128

192.168.10.191

192.168.10.160

192.168.10.223

192.168.10.192

192.168.10.255

192.168.10.224

∴ Valid host :

192.168.10.01 - 192.168.10.30

192.168.10.33 - 192.168.10.62

192.168.10.65 - 192.168.10.94

192.168.10.97 - 192.168.10.126

192.168.10.129 - 192.168.10.158

192.168.10.161 - 192.168.10.190

192.168.10.193 - 192.168.10.222

192.168.10.225 - 192.168.10.254

2018-Spring

3.6> A multinational company is granted a site IP: 172.160.0.15.
Design an IP table with its subnets.

sof Given IP: 172.160.0.15

In Binary mode: 10101100.10100000.00000000.00001111

Since the first 2 bits of network ID is 10, so the given IP address is class B IP.

i.e. 172.160.0.15/16

Here, 16 means that the subnet mask has 16 bits 1s and 16 bits 0s. So the subnet mask in binary mode is:

~~172.160~~

11111111.11111111.00000000.00000000

IP in binary mode,

10101100.10100000.00000000.00001111

After performing AND operation with IP with mask,

10101100.10100000.00000000.00000000

∴ there is no any subnet mask.

∴ Network Address: 172.160.0.0

∴ Subnet mask: 172.160.0.0

1. Design a network for 5 departments containing 29, 14, 15, 23 and 5 computers. Take a network example of IP: 202.83.54.91/25.

Sol: Given:-

IP: 202.83.54.91/25

In binary mode:-

IP: (11001010, 01010011, 00110110, 01011011) /25

Here, 25 means that the subnet mask has 25 bits 1s and 7 bits 0s. So the subnet mask in binary mode is:

11111111, 11111111, 11111111, 10000000

Now, Performing AND operation, IP with mask then,

11001010, 01010011, 00110110, 00000000

i.e. 202.83.54.0

∴ Network Address: 202.83.54.0

We know,

Max. no. of hosts = $2^{\text{no. of host bits}} - 2$

i> for 29 computers;

$$29 = 2^{\text{no. of host bits}} - 2$$

$$\Rightarrow 31 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 4.9 \approx 5$$

So the new subnet mask becomes,

11111111. 11111111. 11111111. 11100000 / 27

i.e. 255. 255. 255. 224 / 27

∴ Subnet Id = ~~256~~ - 224 = 32

∴ Range = difference of 32 = 0, 32, 64, 96, - - - - 224

∴ Network Address: 202.83.54.0

∴ Broadcast Address: 202.83.54.31

∴ Usable host Range = 202.83.54.1 - 202.83.54.30

ii) For 23 Computers:

$23 = 2^{\text{no. of host bits}} - 2$

⇒ $25 = 2^{\text{no. of host bits}}$

∴ no. of host bits = $4.6 \approx 5$

so the new subnet mask becomes,

11111111. 11111111. 11111111. 11100000 / 27

i.e. 255. 255. 255. 224

∴ Subnet Id = 256 - 224 = 32

∴ Range = difference of 32 = 0, 32, 64, - - - - - 224

∴ Network Address = 202.83.54.32

∴ Broadcast Address = 202.83.54.63

∴ Usable host Range = 202.83.54.33 - 202.83.54.62

iii> For 15 Computers:

$$15 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 17 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 4.08 \approx 4$$

So the new ~~mask~~ subnet mask becomes

$$11111111.11111111.11111111.11110000 / 28$$

$$\text{i.e. } 255.255.255.240 / 28$$

$$\therefore \text{Subnet Id} = 256 - 240 = 16$$

$$\therefore \text{Range} = \text{difference of } 16 = 0, 16, 32, 48, 64, \dots, 240$$

$$\therefore \text{Network Address} : 202.83.54.64$$

$$\therefore \text{Broadcast Address} : 202.83.54.79$$

$$\therefore \text{Usable Host Range} : 202.83.54.65 - 202.83.54.78$$

iv> For 14 Computers:

$$14 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 14 + 2 = 16 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 4$$

So the new subnet mask becomes,

$$11111111.11111111.11111111.11110000 / 28$$

$$\text{i.e. } 255.255.255.240 / 28$$

- ∴ Subnet Id = $256 - 240 = 16$
- ∴ Range = difference of 16 = 0, 16, ----- 64, 80, 96, 112, ----- 240
- ∴ Network Address: 202.83.54.80
- ∴ Broadcast Address: 202.83.54.95
- ∴ Usable Host Range: 202.83.54.81 - 202.83.54.94

iv) For 5 computers:

$$5 = 2^{\text{no. of host bits}} - 2$$

$$\Rightarrow 7 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 2.8 \approx 3$$

So the new ~~mask~~ becomes, subnet mask becomes,

$$11111111 \cdot 11111111 \cdot 11111111 \cdot 11110000 / 29$$

i.e. 255.255.255.248 / 29

- ∴ Subnet Id = $256 - 248 = 8$
- ∴ Range = difference of 8 = 0, 8, ----- 96, 112, ----- 248
- ∴ Network Address: 202.83.54.96
- ∴ Broadcast Address: 202.83.54.103
- ∴ Usable Host Range = 202.83.54.97 - 202.83.54.102

2) A large number of consecutive IP addresses are available at 202.70.64.0/19. Suppose that four organizations A, B, C and D request 100, 500, 800 and 400 address respectively, how the subnetting can be performed so that address wastage will be minimum?

So Given:

IP Address: 202.70.64.0 /19

In binary mode: 11001010.01000110.10000000.00000000

Here, 19 means that the subnet mask has 19 bits 1s and 13 bits 0s. So the ~~new~~ subnet mask in binary mode is:

11111111.11111111.11100000.00000000

After performing AND operation IP with subnet mask then,

11001010.01000110.10000000.00000000

i.e. 202.70.64.0

∴ Network Address = 202.70.64.0

We know,

Maximum no. of hosts = $2^{\text{no. of host bits}} - 2$

i) For C (800) hosts:

$$800 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 802 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 9.64 \approx 10$$

so the new subnet mask becomes,

$$11111111.11111111.11111100.00000000 / 22$$

$$255.255.252.0 / 22$$

$$\therefore \text{Subnet ID: } 256 - 252 = 4$$

$$\therefore \text{Range} = \text{difference of } 4 = 0, 4, \dots, 248, 252.$$

$$\therefore \text{Network Address: } 202.70.64.0$$

$$\therefore \text{Broadcast Address: } 202.70.67.255$$

$$\therefore \text{Usable Host Range: } 202.70.64.1 - 202.70.67.254$$

ii) For B (500) hosts:

$$500 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 502 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 8.9 \approx 9$$

so the new subnet mask becomes,

$$11111111.11111111.11111110.00000000 / 23$$

$$\text{i.e. } 255.255.254.0 / 23$$

$$\therefore \text{Subnet ID: } 256 - 254 = 2$$

$$\therefore \text{Range} = \text{difference of } 2 = 0, 2, 4, \dots, 250, 252, 254.$$

$$\therefore \text{Network Address: } 202.70.68.0$$

$$\therefore \text{Broadcast Address: } 202.70.69.255$$

$$\therefore \text{Usable Host Range: } 202.70.68.1 - 202.70.69.254$$

iii) For D(400) hosts:

$$400 = 2^{\text{no. of host bits}} - 2$$

$$\Rightarrow 402 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 8.6 \approx 9$$

So the new ~~max~~ subnet mask becomes,

$$11111111.11111111.11111110.00000000 / 23$$

$$\text{i.e. } 255.255.254.0 / 23$$

$$\therefore \text{Subnet ID} = 202.256 - 254 = 2$$

$$\therefore \text{Range} = \text{difference of } 2 = 0, 2, \dots, 250, 252, 254.$$

$$\therefore \text{Network Address} : 202.70.70.0$$

$$\therefore \text{Broadcast Address} : 202.70.71.255$$

$$\therefore \text{Usable Host Range} = 202.70.70.1 - 202.70.71.254$$

iv) For A(100) hosts:

$$100 = 2^{\text{no. of host bits}} - 2$$

$$\Rightarrow 102 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 6.67 \approx 7$$

So the new subnet mask becomes

$$11111111.11111111.11111111.10000000 / 25$$

$$\text{i.e. } 255.255.255.128 / 25$$

∴ subnet ID = $256 - 128 = 128$

∴ Range = difference of 128 = 0, 128

∴ Network Address: 202.70.72.0

∴ Broadcast Address: 202.70.72.127

∴ Usable Host Range: 202.70.72.1 - 202.70.72.126

Organization	Network Address	Usable Host Range	Broadcast Address	Subnet Mask
C(800)	202.70.64.0	202.70.64.1 - 202.70.67.254	202.70.67.255	255.255.252.0
B(500)	202.70.68.0	202.70.68.1 - 202.70.69.254	202.70.69.255	255.255.254.0
D(400)	202.70.70.0	202.70.70.1 - 202.70.71.254	202.70.71.255	255.255.254.0
A(100)	202.70.72.0	202.70.72.1 - 202.70.72.126	202.70.72.127	255.255.255.128

2023 Fall

Pre

4. a) NAST college is planning to design 3 different department for BE Computer, BCA, BBA, BE Civil, Exam, Research, Account and Library. Explain the process how will you allocate the IP address and subnet for each of the departments using above. Each department consist

So ~~Given~~: of 30 computer with IP Address 192.168.218.0

So IP Address:

Given:

IP Address: 192.168.218.0

In Binary mode,

11000000.10101000.11011010.00000000

Since the first three bits of Network ID is 110. So it class C addressing. Then,

11000000.10101000.11011010.00000000 / 24

i.e. 192.168.218.0 / 24

Here, 8 means that the subnet mask has 24 bits 1s and 8 bits 0s. so the bin subnet mask in binary mode is:

11111111.11111111.11111111.00000000

i.e. 255.255.255.0

After performing AND operation IP Address with subnet mask, we get;

11000000.10101000.11011010.00000000

∴ Network Address = 192.168.218.0

since Each department consists of 30 Computers. Then

Max. no. of host ~~bits~~ = $2^{\text{no. of host bits}} - 2$

$$\Rightarrow 30 = 2^{\text{no. of host bits}} - 2$$

$$\Rightarrow 302 = 2^{\text{no. of host bits}}$$

⇒ no. of host bits = ~~302~~ $\log_2(302)$

$\therefore$ no. of host bits = 5

so the new ~~max~~ subnet mask becomes,

11111111-11111111-11111111-111000000/28

i.e. 255. 255. 255. 224

i) ~~For BE Computer;~~

$\therefore \text{subnet Id} = 256 - 224 = 32$

$\therefore$ Range = difference of 32 = 0, 32, 64, 96, 128, 160, 192, 224, ~~256~~

i) For BE Computer:

Network Address: 192.168.218.0

Broadcast Address: 192.168.218.31

Usable Host Range: 192.168.218.1 - 192.168.218.30

ii) For BCA:

Network Address: 192.168.218.32

Broadcast Address: 192.168.218.63

Usable Host Range: 192.168.218.33 - 192.168.218.62

iii> For BBA:

Network Address: 192.168.218.64

Broadcast Address: 192.168.218.95

Usable Host Range: 192.168.218.65 - 192.168.218.94

iv> For BE Civil:

Network Address: 192.168.218.96

Broadcast Address: 192.168.218.127

Usable Host Range: 192.168.218.97 - 192.168.218.126

v> Exam:

Network Address: 192.168.218.128

Broadcast Address: 192.168.218.159

Usable Host Range: 192.168.218.129 - 192.168.218.158

vi> Research:

Network Address: 192.168.218.160

Broadcast Address: 192.168.218.191

Usable Host Range: 192.168.218.161 - 192.168.218.190

vii> Account:

Network Address: 192.168.218.192

Broadcast Address: 192.168.218.223

Usable Host Range: 192.168.218.193 - 192.168.218.222

Python

viii> Library :

- Network Address: 192.168.218.224

Broadcast Address: 192.168.218.255

Usable Host Range: 192.168.218.225-192.168.218.254.

Department	Network Address	Usable Host Range	Broadcast Address	Subnet Mask
BE Computer	192.168.218.0	192.168.218.01 - 192.168.218.30	192.168.218.31	255.255.255.224
BCA	192.168.218.32	192.168.218.33 - 192.168.218.62	192.168.218.63	255.255.255.224
BBA	192.168.218.64	192.168.218.65 - 192.168.218.94	192.168.218.95	255.255.255.224
BE Civil	192.168.218.96	192.168.218.97 - 192.168.218.126	192.168.218.127	255.255.255.224
Exam Research	192.168.218.128	192.168.218.129 - 192.168.218.158	192.168.218.159	255.255.255.224
Research	192.168.218.160	192.168.218.161 - 192.168.218.190	192.168.218.191	255.255.255.224
Account	192.168.218.192	192.168.218.193 - 192.168.218.222	192.168.218.223	255.255.255.224
Library	192.168.218.224	192.168.218.225 - 192.168.218.254	192.168.218.255	255.255.255.224

2021-Fall

3.6) A company has 3 different departments with 65, 32 and 12 network devices. Explain how you will design network for this company from provided network of 10.10.100.0/24. Provide network address, broadcast address, subnet mask, wildcard mask and Usable IP pool for each subnet;

sg Given:

IP Address: 10.10.100.0 / 24

In Binary: 00001010.00001010.01100100.00000000 / 24

Here 24 means that the subnet mask has 24 bits 1s and 8 bits 0s. So the subnet mask in binary is:

11111111.11111111.11111111.00000000

i.e. 255.255.255.0

Now, Performing AND operation IP with subnet mask.
Then,

00001010.00001010.01100100.00000000

i.e. 10.10.100.0

∴ Network Address: 10.10.100.0

We know,

Max. no. of hosts = $2^{\text{no. of host bits}} - 2$

i) For 65 hosts:

$$65 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 67 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 6.0 \approx 6$$

So the new subnet mask becomes

11111111.11111111.11111111.11000000

i.e. 255.255.255.192

$$\therefore \text{Network Subnet ID} = 256 - 192 = 64$$

$$\therefore \text{Range} = \text{difference of } 64 = 0, 64, \dots, 192$$

$$\therefore \text{Network Address: } 10.10.100.0$$

$$\therefore \text{Broadcast Address: } 10.10.100.63$$

$$\therefore \text{Usable IP Pool: } 10.10.100.1 - 10.10.100.62$$

$$\therefore \text{Wildcard mask} = 0.0.0.192$$

ii) For 32 hosts:

$$32 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 34 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 5.08 \approx 5$$

So the new subnet mask becomes

11111111.11111111.11111111.11100000

i.e. 255.255.255.224

- ∴ Subnet Id : $256 - 224 = 32$
- ∴ Range = difference of 32 = 0, 32, 64, 96, --- 224
- ∴ Network Address : 10.10.100.64
- ∴ Broadcast Address : 10.10.100.95
- ∴ Usable IP Pool : 10.10.100.65 - 10.10.100.94
- ∴ Wildcard mask : 0.0.0.224

iii) For 12 hosts:

$$12 = 2^{\text{no. of host bits} - 2}$$

$$\Rightarrow 14 = 2^{\text{no. of host bits}}$$

$$\therefore \text{no. of host bits} = 3.8 \approx 4$$

∴ the new subnet mask becomes,

11111111.11111111.11111111.11110000

i.e. 255.255.255.240

- ∴ Subnet Id : $256 - 240 = 16$
- ∴ Range = difference of 16 = 0, 16, 32, 64, 80, 96, --- 240
- ∴ Network Address : 10.10.100.96
- ∴ Broadcast Address : 10.10.100.111
- ∴ Usable Host IP Pool : 10.10.100.97 - 10.10.100.110
- ∴ Wildcard mask : 0.0.0.240

Hamming code:

011010001

Q. A seven bits Hamming code is received as 1110111. What is the correct code?

eg: Here, D_7 D_6 D_5 P_4 D_3 P_2 P_1

1	1	1	0	1	1	1
---	---	---	---	---	---	---

step 1: For checking parity bit P_1 , use check 1, ~~check 2~~ bit, ~~check~~ skip 1 bit method. Then, we get;

$D_7 D_5 D_3 P_1 : 1111$

The no. of 1 is even, so we write the value of P_1 as 0. This means no error.

step 2: For checking parity bit P_2 , use check 2 bit, skip 2 bit method. Then, we get;

$D_7 D_6 D_3 D_2 : 1111$

The no. of 1 is even, so we write the value of P_2 as 0. This means no error.

step 3: For checking parity bit P_4 , use check 4 bit, skip 4 bit method. Then, we get;

$D_7 D_6 D_5 P_4 = 1110$

The no. of 1 is odd, so we write the value of P_4 as 1. This means error exists.

The error word, $E =$

P_4	P_2	P_1	0	0	1
1	0	0	P_1	P_2	P_3

The decimal equivalent of E is ~~4~~ 1.

Hence, the ~~4th~~ bit in the code word B incorrect.

For correction, Flip 1st bit 1 as $1 \rightarrow 0$, which gives the correct code:

0110111.

3. a) A bit stream 10011101 is transmitted using the standard CRC. The generator polynomial is x^3+1 . Show the actual bit string transmitted. Suppose the third bit from left is inverted during transmission. Show that this error is detected at receiver side.

So Here,

Given bit stream: 10011101

A generator polynomial: $x^3+1 = 1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0$

For Sender: i.e. 1001

Since the generator polynomial is encoded as 1001. So we append $(4-1) = 3$ 0s to the bit stream as

10011101000

Now, Divide the bit stream by 1001 as

1001) 10011101000 (10001100

$$\begin{array}{r}
 1001 \downarrow \\
 \times 0001 \\
 \hline
 0000 \downarrow \\
 \times 0011 \\
 \hline
 0000 \downarrow \\
 \times 0110 \\
 \hline
 0000 \downarrow \\
 \times 1101 \\
 \hline
 1001 \downarrow \\
 \times 1000 \\
 \hline
 1001 \downarrow \\
 \times 0010 \\
 \hline
 0000 \downarrow \\
 \times 0100 \\
 \hline
 0000 \\
 \times 100 \leftarrow \text{CRC}
 \end{array}$$

~~For rec~~

Hence the CRC bit is 100. Now ~~Appen Rep~~ Append this CRC to the original bit stream as

10011101100

For receiver,

If we invert the ~~the~~ third bit from left then bit stream becomes as

10111101100

Now, Divide this bit stream by 1001.

1001) 10111101100 (10101000

$$\begin{array}{r} 1001 \downarrow \\ \times 0101 \\ \hline 0000 \\ \times 1011 \\ \hline 1001 \\ \times 0100 \\ \hline 0000 \\ \times 1001 \\ \hline 1001 \\ \times 0001 \\ \hline 0000 \\ \times 0010 \\ \hline 0000 \\ \times 0100 \\ \hline 0100 \\ \times 100 \\ \hline 0100 \end{array}$$

← Remainder

In remainder there is none zero value. This showed that the error is detected at the receiver side.

Bit stuffing Sol

2018-Fall

34

Q.6> A bit string 011011111110011111011111111100000 needs to be transmitted at the data link layer, what is string actually transmitted after bit stuffing, if pattern is 01111110?

sol:
Given bit string: 011011111110011111011111111100000
Flag pattern: 01111110

When five consecutive 1s are detected in the data, a 0 is inserted after them to avoid confusion with the flag pattern.

After bit stuffing, the data becomes:

~~011011111010011111011111100000 011111100000~~

01101111101001111101111101111110000000

Ans